Seunghyun Yoo

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Research Interests

Mechanistic Interpretability, Technical AI Safety, and Multi-Agent Systems

Education

Purdue University Aug. 2024 – Expected Dec. 2027
B.S. in Artificial Intelligence; Minor in Mathematics *West Lafayette, IN*

Publications

- **Seunghyun Yoo**. *White Box Evidence Packages for Policy Audit Reports*. Second Workshop on Technical AI Governance Research (TAIGR), ICML 2026. Sole author. PAPER | POSTER
- Meliksah Yorulmazlar, Justice Sefas, **Seunghyun Yoo**, and Lulu Shao. *Feature Evolution in Diffusion Language Models*. Non-Autoregressive Language Models Workshop, COLM 2026. Poster. PAPER | POSTER | LIVE DEMO

Research Experience

Research Fellow Sep. 2026 – Present
Supervised Program for Alignment Research; Mentors: Yuxiao Li and Di Wu *Remote*

- Studying whether reading the internal states of interacting agents reveals influence that transcripts miss, in *Who's Steering Whom? Interpretable Influence and Equilibria in Human-Agent Systems*.
- Designed and ran pre-registered steer-and-read pilots on Qwen2.5 models: agent A is steered along a contrastive activation direction while agent B's residual-stream projection is read turn by turn.
- On 300 GSM8K problems, a probe on B's pre-answer activations predicted when B would adopt A's wrong answer (AUROC 0.73), outperforming an LLM judge reading the same transcript (0.56).

Undergraduate Researcher (Team Lead) Jun. 2025 – Present
Governance and Responsible AI Lab (GRAIL) *Purdue University*

- Designed a policy audit evaluation across 60 AGORA cases, 600 reports, and ten evidence conditions, showing that audit reports can cite irrelevant internal model evidence while still reading as plausible; authored the resulting TAIGR paper.
- Built a policy representation analysis in Gemma 2 2B, from corpus preparation through SAE extraction, statistical testing, and causal interventions; identified 35 layer-24 features stable across five permutation seeds in 2,438 records.
- Developed AGORA-RAGTruth, a policy hallucination benchmark with manual span annotations and governance tags, with mechanistic analyses separating retrieved-context use from parametric knowledge.
- Annotated and validated AGORA policy documents and redesigned the Explore-AGORA graph schema linking policy passages to risks, strategies, and institutional metadata.

Research Fellow Feb. 2026 – Jun. 2026
Algoverse AI Safety Fellowship; Supervisor: Justice Sefas *Remote*

- Analyzed SAE features in Dream-v0-Base-7B across six layers and 64 denoising steps, covering activation statistics, feature interpretation, and token-commitment visualizations.
- Identified recurring feature patterns: tracking masked tokens, activating near token commitment, and responding selectively to unmasked tokens.
- Compared token unmasking across math, natural-language, and code reasoning prompts; the resulting paper, *Feature Evolution in Diffusion Language Models*, was accepted to the Non-Autoregressive Language Models Workshop at COLM 2026.

Hackathon Projects

Ethogram LIVE DEMO | CODE Oct. 2026
AI Village × Grove Research AI Swarm Dynamics Hackathon

- Built a pipeline that turns multi-agent logs into a plain-language behavior dictionary, a storyline of phases and turning points, and an interactive behavior atlas, with every claim linked to the events behind it.
- Validated behaviors with two independent LLM judges and blind detection by a different model; two-judge agreement reached $\kappa = 0.89$ and 0.86 on records of 20,932 and 67,969 events.

MARGIN Arena: Agents Under Resource Limits PAPER | CODE Sep. 2026
AI Incident Response Hackathon; Apart Research *Remote*

- Designed and published a Python environment for studying agent behavior under five independently configurable limits: tokens, time, verification calls, tool calls, and sessions.
- Integrated Inspect and ControlArena tasks with original scoring, full execution records, and optional activation capture for interpretability; validated resource accounting and stopping behavior with 50 software tests.

Denoising Trajectories as Early-Warning Signals for Diffusion Language Model Control PAPER | CODE Mar. 2026
AI Control Hackathon; Apart Research / Redwood Research *Remote*

- Built SAE-based monitoring and steering for LLaDA-8B-Instruct, with executable shell benchmarks and tests under prompt and payload variation.
- Trajectory monitoring detected attacks at mean denoising step 9.60, versus 63.00 for output-only checks; early detection held under variation (10.53 versus 63.00).

Teaching

AI Study Leader Sep. 2025 – Present
LikeLion US / OZ Lab *West Lafayette, IN*

- Teach weekly sessions on LLM From Scratch, MiniArena, and Interpretability for Purdue's Korean student community.

Selected Posters and Presentations

- *Co-CoT: A Prompt-Based Framework for Collaborative Chain-of-Thought Reasoning*. RISE AI Conference, Oct. 2025. Poster and presentation; first author.
- *Factually Grounded and Human-Aligned RAG Systems for AI Governance and Policy*. Purdue URC, 2025. Poster.
- Internal GRAIL presentations: *Explore-AGORA 2* (Feb. 2026) and *DPO + Explore-AGORA* (Nov. 2025).

Honors and Recognition

- Full Member, Sigma Xi, The Scientific Research Honor Society, Jul. 2026.
- Travel Grant, Grove Research, 2026 ($1,500).
- Travel Grant, TAIGR Workshop at ICML 2026 ($1,000).
- Travel Grant, RISE AI Conference, University of Notre Dame, Oct. 2025.
- Purdue PLaCE Student Spotlight, 2024.

Technical Skills

Programming and ML	Python, PyTorch, Hugging Face Transformers, Git
Language models	RAG, diffusion models, model evaluation
Interpretability	Sparse autoencoders, activation patching, probing, TransformerLens, NNsight, NDIF

Activities

AI Safety Purdue and ML@Purdue, general member; Purdue eSports, Overwatch Junior Varsity.